EFY TESTED
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Contactless Hand Sanitiser Dispensing System

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Hand sanitisation is important to stop spreading Covid-19 and other transmissible diseases. Even touching the container spout is unhygienic, so it is better to deploy an auto-dispenser to sanitise your hands without touching anything except the sanitising liquid. It can be useful especially for industrial units, workplaces, hospitals, and shopping centres, etc.

But most conventional contactless sanitisers use an LDR and infrared (IR) sensors, which frequently malfunction due to the ambient sunlight and high ambient, etc. So here is an auto-dispenser that uses ultrasonic sensing instead to dispense the sanitiser.

The dispenser uses ultrasonic sensor HC-SR04 to detect hands when these are put beneath the container. It dispenses the required amount of sanitiser automatically for specific time and then gets ready for the next action within half a second. The

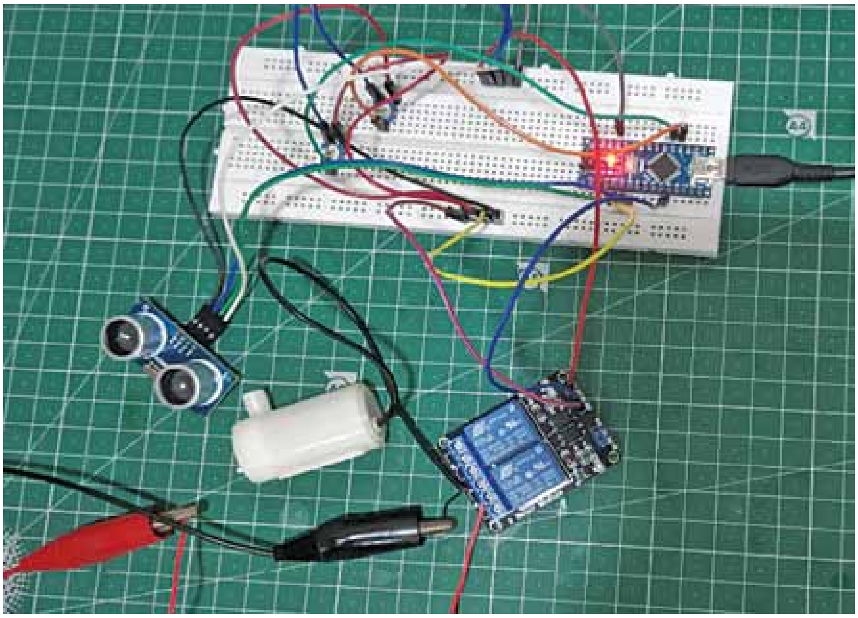

Fig. 1: Author's prototype

Table 1: Ultrasonic Sensor Specifications

Name	Specifications
Operating Voltage (DC)	5V
Operating Current	15mA
Operating Frequency	40kHz
Maximum Range	4 metres
Minimum Range	2cm
Accuracy	3mm
Measuring Angle	15 degrees
Trigger Input Signal	10μS TTL pulse
Dimensions	45mm×20mm×15mm

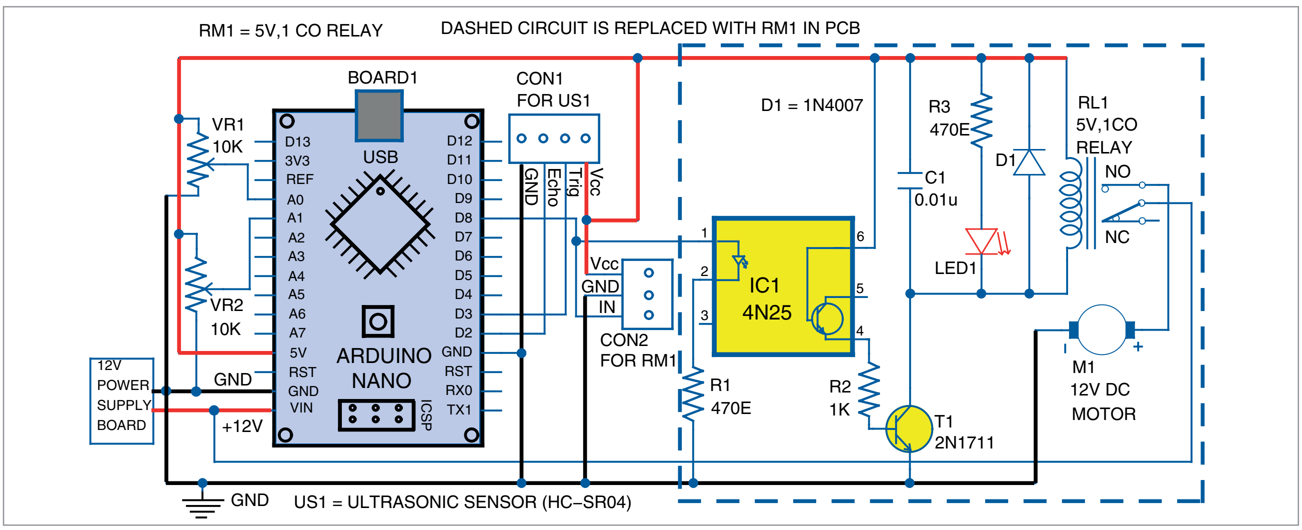

Fig. 2: Circuit diagram